Report

Student at Risk

Omar A. shakir Al-Faiadh 2017351677
Shamsa Ahmed Al Nasri 201201405
Marian Al Nuaimi 201708147
Mohamed Zoghbour 202050082
Muna Ali Al-Shamsi- 700045181

College of Business and Economics, United Arab Emirates University

BANA 600 Business Analytics Applications (01)

Prof. Omar Hujran

September 29, 2024

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Introduction and Background:

1. Overview

The landscape of Higher Educations has transformed rapidly, characterized by the increase of students' demographic and increasing academic standards complexity. In response to these changes, many of Higher Education institutes have embraced different approaches and strategies, utilizing analytical methodologies for better understanding of the variety students' demographic and academic characteristics insights to enhance the alignment between the institution practices and its vision, mission and the strategic objectives.

The demographic characteristics of students encompass various dimensions, and one of the characteristics is known as "Students at Risk". The concept is considered a main and critical issue that faces Higher Education institutes, giving the significance of this concept in attainment of students' intended learning outcomes and graduate attributes. As a result, many strategies have been developed to mitigate the impact of this issue.

The concept "Students at Risk" refers to students who have higher likelihood of not persisting in their academic program (O'Leary, 2020). Due to a combination of the Students at Risk characteristics, the awareness of the concept's dimensions became essential to utilizing and leveraging analytical methodology relying on new approaches that define the attributes that may lead to having students at risk and supports monitoring students academically.

Students' data and student characteristics are considered very significant data that was used in performing statistical analysis to identify the key attributes that contribute to classifying the students as "at risk", predicting their future performance and predicting the new students' possibility of being classified at the same classification based on the defined attributes. The efficiency of the new techniques such as Machine Learning (ML), has demonstrated high predictive accuracy, and provides a detailed understanding of these attributes which enhances decision making and contributes to mitigating the issue.

2. Project Scope

The project is aimed to explore and identify the main attributes that contribute to having students at risk by using ML and use that information to predict the new students who might be at risk.

3. Problem Statement

Understanding the challenges faced by students at risk is essential, however the higher education institutes struggle to identify the early students at risk due to the lack of an accurate predictive model. This gap limits the implementation of effective intervention that is aimed at improving the student's success.

The project aims to use different ML models to address this issue through identifying the key attributes associated with at-risk students. Also, the project seeks to enhance the institution's capability to implement proactive strategies.

Literature Review

The literature review for this project will explore multiple forms of traditional statistical methods, supervised and unsupervised machine learning models. And conclude with a summary of the models/methods, and the common elements between them.

1. Traditional Statistics:

Traditional statistics existed before computers. It relies on data summarization and analysis and is usually used on smaller sets of data where an assumption about the data is needed before the work starts. Despite the age of this approach and the availability of more advanced methods, it is still one of the most widely used methods to forecast (ToolsGroup, n.d.). In this section we will explore a few studies that have approached the topic of Students at Risk using the traditional statistical methods of hypothesis testing and regression.

1.1 Hypothesis Testing:

Hypothesis Testing is a traditional statistical method that is concerned with explaining assumed theories based on certain IF-statement parameters (Harvard Business School Online, 2021). In their work, Gallego et al (2021) aimed to identify a profile of first year health science students in cognitive abilities and learning strategies instruments by using tests such as Pearson's correlations and $\chi 2$-square test on the results obtained from different questionnaires. They divided the 284 students into three clusters, cluster-1 were students who scored high in cognitive abilities learning strategies instruments, cluster-2 were students who scored high in cognitive abilities and low in learning strategies instruments. And the students in cluster-3 scored low in cognitive abilities and low in learning strategies instruments and this resulted in finding a correlation between cognitive abilities and learning strategies instruments where lower grades in either aspect resulted in poorer test scores and therefore, increased the potential of more students at risk due to increased stress.

In another study by Abdi Zarrin, T et al. (2014), the researchers measured the fear of failure and self-regulation and its effect on academic procrastination in 198 university students across five programs. They also used questionnaires to gather information

from the students and performed a t-test analysis, Pearson's correlations and Linear Regression. The results in the t-test showed a higher score for female students when it came to self-regulation and fear of failure, which in turn means that they scored less when it came to academic procrastination. In Pearson's correlations, procrastination had a positive relationship with fear of failure, while scoring negatively with self-regulation parameters. In the Linear Regression, the researchers used one aspect of self-regulation (responsibility) in their test against procrastination and the results indicated that lower responsibility meant higher procrastination. Moreover, they conducted a second test in which they added fear of failure to responsibility against procrastination, and the results indicated that higher fear of failure and lower responsibility resulted in higher procrastination.

Table 1: Comparison of studies using the Hypothesis Testing technique.

Reference	Technique used	Data used	Validation Method	Results
Gallego, Cobos and Gallego (2021)	Pearson's correlations and χ2-square test	284 health science first year students	Comparison tests	They found a correlation between poor cognitive abilities and poor learning strategies instruments. Lower grades in either resulted in an increased potential of more students at risk due to increased stress.
Abdi Zarrin, T., Gracia, E., & Sullivan, J. R. (2014)	t-test analysis, Pearson's correlations and Linear Regression	198 students from five majors	Stepwise linear regression	They found a negative correlation between procrastination and self-regulation while finding a positive correlation between procrastination

				and fear of failure.

1.2 Linear Regression

Linear Regression is an analysis that can predict the value of a dependent variable using an independent variable (IBM, n.d.). In a study conducted in Universiti Teknologi MARA in Malaysia (Rusli & Ali, 2009), 206 students' performance were evaluated using the cumulative GPA (CGPA) after their first undergraduate semester. The CGPA was then put into three prediction models with the aim to predict the students' final CGPA in their undergraduate studies, and they were Linear Regression, Artificial Neural Network and a Decision Tree. All the models scored over 80% accuracy, but the most accurate one was the artificial neural network model. In their works (Dagdagui, 2022) the researchers tested 521 first year university students in the colleges and universities of the Cordillera Administrative Region with variables including their socioeconomics background, high school background and their admission test scores. The gathered data was tested using a multiple regression model where the results ranged from poor to average in the various variables tested with the main deciding variables of their performance to be their admission test results and high school performance.

Table 2: Comparison of studies using Linear Regression.

Reference	Technique used	Data used	Validation Method	Results
Rusli & Ali, 2009	Linear regression, artificial neural network and a decision tree	206 undergraduate students' CGPA	Cross validation, training, and testing.	All the models scored over 80& accuracy, but the artificial neural network was the mode accurate. Linear regression had the worst outcome.
Dagdagui, 2022	Multiple linear regression	521 undergraduate students in Cordillera Administrative Region	Cross validation, training, and testing.	High school scores and admission test scores were the main variables in predicting the student performance.

2. Machine Learning:

Machine Learning is a subset of artificial intelligence and is largely defined as a machine that is capable of mimicking human thinking. In the 1950s, it was also defined as “the field of study that gives computers the ability to learn without explicitly being programmed” by Arthur Samuel (MIT Sloan School of Management, n.d.). In this section, we will be exploring a few studies that discuss machine learning methods, in both supervised and unsupervised applications.

2.1 Supervised Learning:

Supervised learning is when a model is trained using specific labeled data to accurately predict data it has not seen before through two main paths, regression, and classification (IBM, n.d.). In the study done by Karalar H. et al. (2021), the effects of COVID-19 were tested given its considerable effect on many aspects of daily life, including education. The study was conducted in Turkish state university, on 2116 first year students using classification methods in an ensemble learning approach featuring models like Gradient Boosting, Decision Trees, Random Forest, Logistic Regression and Artificial Neural Networks. This approach resulted in a combination of 91 ensemble models and showed that the combination of Extra Trees, Random Forest and Logistic Regression was the best at predicting Students at Risk with 90.34% accuracy.

Unlike most countries however, German universities do not closely monitor their students’ progress, nor do they put any pressure on them to finish their studies on a timeline, but they still present a risk to the economy due to misuse of resources (Berens et al., 2019). In this study, researchers worked on developing an Early Detection System to help reduce the student attrition by using an ADABoost method that combines Regression Analysis, Neural Network and Decision Trees on 12730 students in the state university and 6297 students in the private university. The first semester had 79% accuracy after the first semester in the state university, and 90% accuracy after the fourth semester. While the first semester in the state university had an 85% accuracy and a 95% accuracy in the private university.

Early detection and prevention are the common elements between the studies so far, and it is also relevant in the next research (Lottering, Hans, & Lall, 2020). A dataset of 4419 students were tested using Random Forest, Support Vector Machines, Decision Trees, Logistic Regression, K-Mearest Neighbor and Naïve Bayes to anticipate those who are at risk before they drop out. The Random Forest model had the highest accuracy amongst the models used with 94.14%.

Table 3: Comparison of studies using Supervised Learning technique.

Reference	Technique used	Data used	Validation Method	Results
Latrellis et al. 2021	Gradient Boosting, Decision Trees,	2116 first year students	Cross-validation	91 ensemble models and showed that

	Random Forest, Logistic Regression, Artificial Neural Networks, Extra Trees and Quadratic Discriminant Analysis.	from a Turkish State university		the combination of Extra Trees, Random Forest and Logistic Regression was the best at predicting Students at Risk with 90.34% accuracy.
Berens et al., 2019	Regression Analysis, Gradient Boosting, Neural Network and Decision Trees	12730 students in the state university and 6297 students in the private university	Cross-validation	The first semester had 79% accuracy of prediction in the state university, and 90% accuracy after the fourth semester. While the first semester in the state university had an 85% accuracy and a 95% accuracy in the private university.
Lottering, Hans, & Lall, 2020	Random Forest, Support Vector Machines, Decision Trees, Logistic Regression, K-Mearest Neighbor and Naïve Bayes	4419 students	Cross-validation	The Random Forest model had the highest accuracy amongst the models used with 94.14%.

2.2 Unsupervised Learning

Unsupervised Learning models are models that manage to find patterns, groupings and classifications in unlabeled datasets without any human guidance (IBM, n.d.). In this

study conducted by Palani et al. (2021), the main concern was gauging of the student engagement as it was difficult for instructors to do so for every student in online or virtual learning environments. The researchers built unsupervised learning models such as Fuzzy C-Means Clustering, Hierarchical Clustering and K-prototype to classify their dataset of 32593 students based on the students' engagement levels. The K-prototype scored the highest in the silhouette coefficient at 75% in comparison to the other models.

In another study conducted by Moreno et al. (2022), the focus was to find out why did the graduate number decrease by at least 10000 between the years of 2019 and 2020, and to find a reasoning of why the total number of graduates was significantly less than the total number of admissions. The researchers used K-Means, DBSCAN and BIRCH in their analysis, in which they found that K-Means was the best clustering method due to the coefficient scores in comparison to the other models.

Table 4: Comparison of studies using Unsupervised Learning technique.

Reference	Technique used	Data used	Validation Method	Results
Palani et al. (2021)	Fuzzy C-Means Clustering, Hierarchical Clustering and K-prototype	32593 students	silhouette coefficient	The K-prototype scored the highest in the silhouette coefficient at 75% in comparison to the other models
Moreno et al. (2022)	K-Means, DBSCAN and BIRCH	117,069 students	silhouette coefficient	that K-Means was the best clustering method due to the coefficient scores in comparison to the other models

3. Summary of the Literature Review:

The literature review explored different data analysis techniques, including traditional statistics, supervised and unsupervised machine learning models. While also including summary tables that compare the methods and models used, and what could possibly be more suitable to analyze a topic such as Students at Risk. It's also evident that a combination between traditional statics and machine learning models had the better

results in the identification and classification of Students at Risk, alongside an analysis of their behavior and personality instead of just relying on academic performance.

Research Methodology and Data Preparation

1. Data Collection

A College of Business and Economics student dataset was collected during Fall 2023. It contains 1729 records and multiple variables related to students' academic performance, demographic information, and educational background. For this analysis we will use the KNIME software. The purpose of this research is to analyze potential factors contributing to students being at risk of academic underperformance. Before applying the analysis, we created a class variable based on the cumulative GPA to classify the students as 0 at-risk with a GPA equal to or less than two and 1 not at-risk with a GPA greater than two.

The dataset includes the following key columns:

- **Semester:** Denotes the academic term during which the data was collected, covering Fall 2023.
- **Gender:** Indicates the student's gender (M for male, F for female).
- **MAjor:** Represents the student's primary area of study, using University abbreviation.
- **Major:** Represents the student's primary area of study, including majors such as Finance & Banking, Business Administration, Economics, Accounting and Statistics & Data Analytics. The minimum required credit hours for graduation is 120 credit hours for all CBE programs, except for Statistics and Data Analytics. Which is at 121 credit hours. (United Arab Emirates University, 2023)
- **Concentration:** In cases where students have a specialization, such as Human Resources Development or Analytics for Business, these fields provide additional context.
- **Minor:** In cases where students opt to declare a minor after completing 30+ credit hours from the current program. A minor program requires 18 credit hours to complete. (United Arab Emirates University, 2023).
- **Cumulative GPA:** Reflects the student's cumulative Grade Point Average, serving as a crucial metric for academic performance, with values ranging from 0.00 to 4.00.
- **Earned Hours:** Total academic hours earned toward degree completion.
- **Expected Graduation Term:** Predicts the student's expected graduation term, which helps track progress towards completing their studies. Numbers ending with 10 (e.g. 202710) indicate the Fall semester and numbers ending with 20 (e.g. 202720) indicate the Spring semester.
- **High School Type:** Categorizes the type of high school the student attended (e.g., Science, General, Advanced).

- **Nationality, City, Emirate:** Provide demographic details of the students, including their nationality, city of residence, and emirate.

This comprehensive dataset will allow us to assess relationships between demographic and academic factors and apply predictive models to identify at-risk students.

2. Data Cleaning

The dataset underwent a thorough cleaning process to ensure it was suitable for further analysis and modeling. Two key aspects were addressed during this stage: missing values and outliers.

Missing values were identified in several fields, particularly in academic performance variables. Appropriate imputation methods were applied to ensure that missing data did not bias the analysis. For categorical fields, missing entries were replaced with NA. However, the new students who still didn't have a cumulative GPA or 0 credit hours were excluded from the analysis.

To calculate the potential graduation term, we have calculated the remaining credit hours based on the previously mentioned hours needed to graduate (120 for all majors except for Statistics at 121 credit hours). And then used that resulting number to calculate the remaining semesters until they graduate by using the minimum required hours for students at the university as our benchmark.

These approaches help preserve the dataset's integrity while reducing the risk of introducing errors. Also, we have noticed that the earned hours variable is not normally distributed, so we used the normalizer node in KNIME software to avoid bias in the data.

3. Exploratory Data Analysis (EDA)

Following the cleaning process, exploratory data analysis (EDA) was conducted to gain an initial understanding of the data's structure and key patterns.

3.1 Descriptive Statistics

Summary statistics were computed for the key numerical variables in the dataset, including Cumulative GPA, Earned Hours, and Expected Graduation Term. For instance, the average cumulative GPA across students was found to be 2.86, with a standard deviation of 0.705, indicating moderate variation in academic performance. Descriptive statistics provided a foundational understanding of the central tendencies and distribution of the data.

3.2 Data Visualization

Several visualizations were generated to provide insights into the distribution of key variables. Histograms were created to visualize the frequency distribution of variables like Cumulative GPA, which appeared normally distributed with a slight right skew. Box plots were used to detect outliers and identify the spread of variables such as Earned Hours and Expected Graduation Term. Correlation heatmaps were also employed to examine relationships between variables, revealing a strong positive correlation between Cumulative GPA and Earned Hours, suggesting that students with higher earned hours tend to maintain higher GPAs.

4. Data Preparation for Modeling

Once the data was cleaned and explored, it was prepared for the modeling phase. The following steps were performed to ensure that the data was suitable for machine learning applications.

4.1 Feature Selection

Key features were selected for the predictive modeling process. Variables such as Cumulative GPA, Earned Hours, and Major were chosen due to their predictive potential. Additionally, categorical variables like Gender and High School Type were included to capture the diversity of the student population. One-hot encoding was applied to categorical variables to convert them into a format suitable for machine learning algorithms.

4.2 Train-Test Split

To evaluate the performance of the predictive models, the dataset was split into training and testing sets. A 70:30 train-test split was applied, where 70% of the data was allocated for model training and 30% was reserved for testing. This ensures that the model can be evaluated on unseen data, preventing overfitting and ensuring the model's generalizability.

5. Machine Learning Models Applied

To predict which students are at risk, several machine learning models were applied to the prepared dataset. Each model was chosen for its ability to handle classification tasks and provide insights into different aspects of student performance. The following models were utilized:

- **Random Forest:** A robust ensemble method that combines multiple decision trees to improve accuracy and mitigate overfitting by averaging the predictions of individual trees.
- **Logistic Regression:** A statistical model widely used for binary classification problems, predicting the likelihood of students being at risk based on their features.

- **Decision Tree:** A straightforward, interpretable model that uses a tree-like structure to make decisions based on data attributes.
- **Support Vector Machine (SVM):** A classification model that finds the optimal boundary (hyperplane) to separate students into different risk categories.
- **Gradient Boosted Trees:** An advanced ensemble technique that builds models sequentially, with each model correcting the errors of the previous ones, resulting in improved prediction accuracy.

To address the class imbalance in the dataset, SMOTE (Synthetic Minority Over-sampling Technique) was employed before each ML model. This technique generated synthetic samples for the underrepresented class, ensuring that the models were trained on a more balanced dataset and, therefore, better equipped to identify students at risk.

Dashboards and Summary Statistics:

Figure 1: Summary Statistics

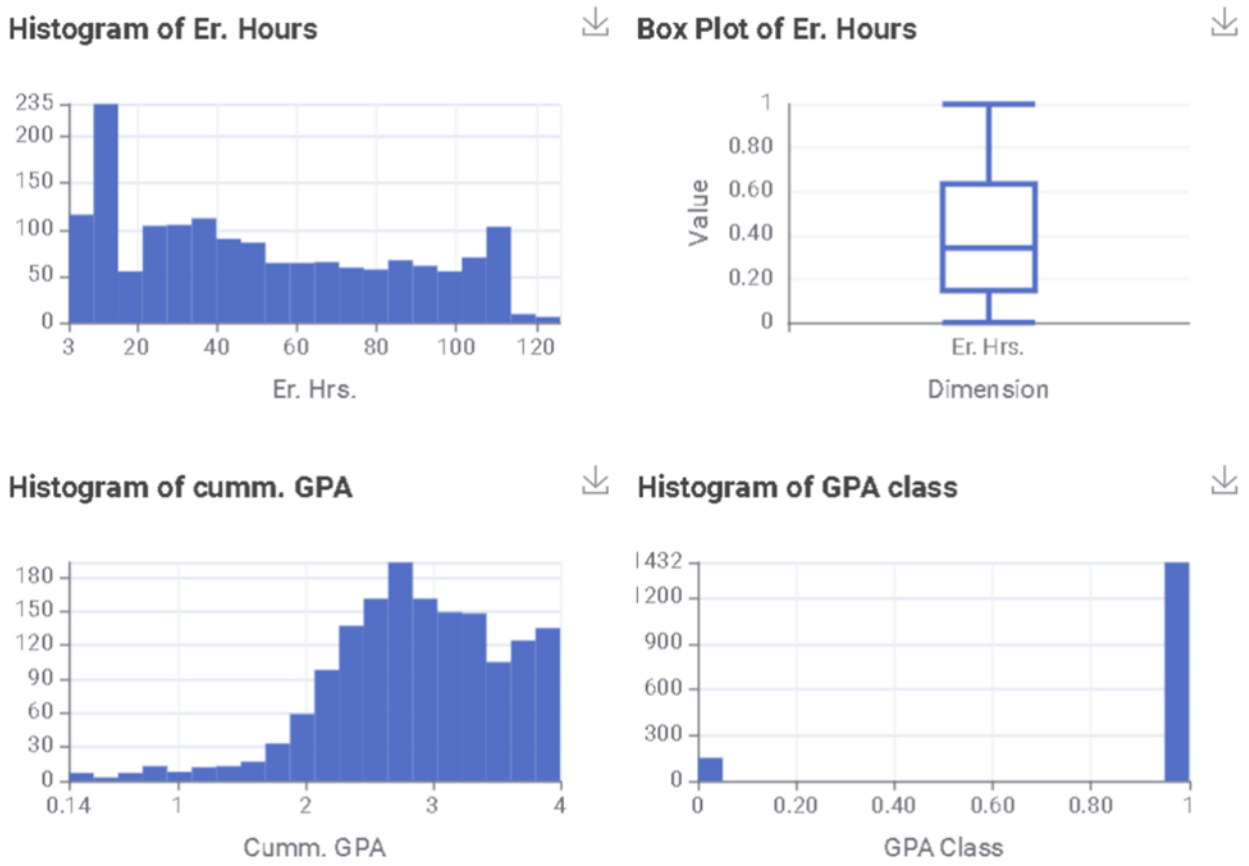


Figure 2: Dashboard of Histograms and Box Plot.

Row ID	S Column	D Min	D Max	D Mean	D Std. de...	D Variance	D Skewness	D Kurtosis	D Overall ...	I No. mis...	I No. NaNs	I No. +oos	I No. -oos	D Median	I Row co...	Histogram
GPA Class	GPA Class	0	1	0.905	0.294	0.086	-2.757	5.61	1,432	0	0	0	0	?	1583	
Cumm. GPA	Cumm. GPA	0.14	4	2.86	0.706	0.498	-0.698	0.861	4,527.34	0	0	0	0	?	1583	
Er. Hrs.	Er. Hrs.	3	126	51.073	33.379	1,114.175	0.362	-1.158	80,849	0	0	0	0	?	1583	

In Figure 1, we can see some visualizations of Histograms and a box plot for the data, The Histogram of GPA class shows that the data is imbalanced, where there are 151 students who are at risk while there are over 1400 students who are not at risk. This might cause some problems with training models. We could use SMOTE node in our model training to cause the problems of imbalanced data. The Histogram of cumulative GPA shows that the cumulative GPA of UAEU students is not normally distributed. This might be caused by the variance of acceptance requirements when it comes to accepting new students or to other factors that we might find out if we investigate this matter.

As Figure 2 shows, the cumulative GPA ranges from 0.14 to 4 with a mean of 2.86 and Standard deviation of 0.706. The skewness of the cumulative GPA is -0.698 which indicates a left skewness as we can see in Figure 1. Figure 1 shows the Histogram of Er. Hours and the Box plot of Er. Hours. The box plot shows that there are no outliers in the Er. Hours and the Histogram shows that Er. Hours is not normally Distributed. Figure 2 shows that Er. Hours range from 3 hours to 126 hours with a mean of 51.073 and standard deviation of 33.379. The skewness of Er. Hours is 0.362, which indicates a right-skewness of the Histogram.

This means that most students are new to the university, and the distribution of the Histogram shows that the students who are taken into consideration in this data have very different credit hours with high variance between each other.

Figure 3: Dashboard of the Dataset.

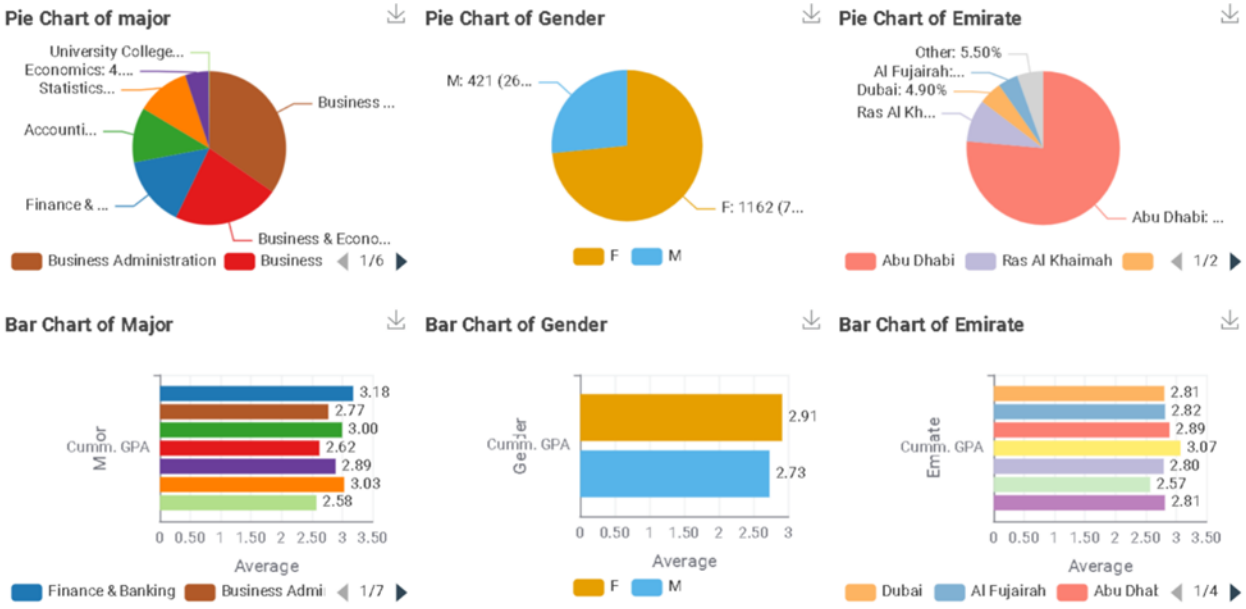


Figure 3 shows some visualizations of Major, Gender, and Emirate. The pie charts in this dashboard show the percentage of each value's frequency while the bar charts show the Average cumulative GPA of these values. The first factor we'll discuss is the Major. As shown in the Pie chart of Major, Business administration is the most common major in our data with 34.62% enrolled students from the dataset. The second most frequent major is Business and Economics with 22.55% of the students. Then, Finance and Banking comes as the third most common major among students with 14.84%. 11.56% of students have Accounting as their major and Statistics majors are 11.31% of the students in this data. Economics students are at 4.99% and lastly, University College Program is the least common major in this data where only 0.13% of students enrolled in it.

The Bar Chart of Major shows the average cumulative GPA of all these majors. Finance and Banking students have the highest Average Cumulative GPA among all other majors while statistics and data analytics came second with an average cumulative GPA of 3.03. Accounting students came third with an average Cumulative GPA of 3.00 and Economics students had the fourth highest cumulative GPA among all majors with 2.89 average cumulative GPA. Then, Business Administration students and Business & Economics students came next, where they had an average Cumulative GPA of 2.77 and 2.62 respectively. The major with the least average Cumulative GPA was University College Program students with 2.58 Average cumulative GPA.

To have some simple statistics about Gender, we can look at the bar chart of gender and the pie chart of gender in Figure 3. The pie chart of gender shows that 73.4% of students are female while only 26.6% of students are male. The bar chart of Gender shows the average cumulative GPA of both female and male students. The female students have higher average cumulative GPA than male students, where female students' average cumulative GPA was 2.91 and male students' average cumulative GPA was 2.73.

The dashboard in Figure 3 also shows a pie chart of Emirate and a bar chart of it as well. The pie chart shows that 76.41% of students are from Abu Dhabi Emirate, which is the Emirate with the highest number of students. The second highest Emirate that students are from is Ras Al Khaimah with 8.95% of the students being from it. After that, Dubai comes third and Al Fujairah comes fourth with 4.9% of students and 4.24% of students respectively. The students from the remaining Emirates of Sharjah, Ajman and Umm Al Quwain totaled the remaining 5.5% of the students.

The bar chart of Emirate shows that the students of Sharjah had the highest Average cumulative GPA among all Emirates where they had an average cumulative GPA of 3.07. Students from Abu Dhabi Emirate had the second highest average cumulative GPA of 2.89. Al Fujairah came as the third highest average cumulative GPA of 2.82. Both Dubai and Ajman came equally with an average cumulative GPA of 2.81. Followed by Ras Al Khaimah with a close average cumulative GPA to the ones before, at 2.8. Lastly, students from Umm Al Quwain had the lowest average cumulative GPA of 2.57, which is quite the difference in comparison to the other emirates.

Models:

1. Decision Tree Model:

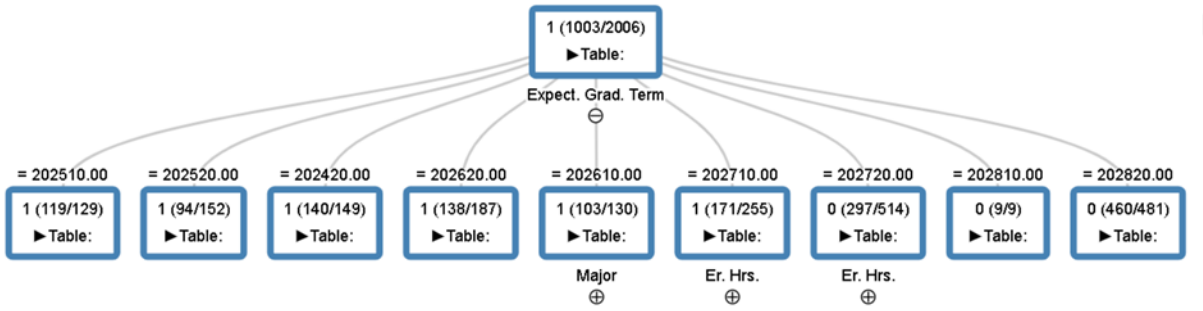
One of the predictive models that we can use in identifying whether the student would be expected to be at risk or not is the decision tree model.

This model is a good choice to test because the target variable is a binary variable, the student at risk or not at risk, and most of the independent variables are categorical. In our case, we used partitioning to split the data into two parts, the first part was set to be 70% and the second part was set to be 30%. The split was done using a random drawn method. For the first part, SMOTE, an oversample minority classes method, was used to fix the problems that might happen because of the imbalanced data we have.

Afterwards, the first part was used for training the model. This model first tries to find out which features, depending on the independent variables, are the most significant in separating or splitting the samples that're 0s and the ones that're 1s. In deciding what the features of the decision tree would be, the model in our case would apply a multiple way splitting for the categorical data and binary splitting for the numerical data.

Afterwards, the model would evaluate all categorical and numerical variables for all potential splits or possible features to be used as branches. Usually, gini impurity is used in this model for evaluation. For our model, we set GPA class (0 for students at risk and 1 for students not at risk) as the binary variable and evaluated the independent variables to choose the features for the decision tree. The trained model evaluated the independent variables and selected the best features to be taken into consideration in the multiple way split of the branches. MDL method was used in training the model to avoid overfitting problems, considering that this method ensures that the number of threads in the decision tree will not be too high and cause overfitting.

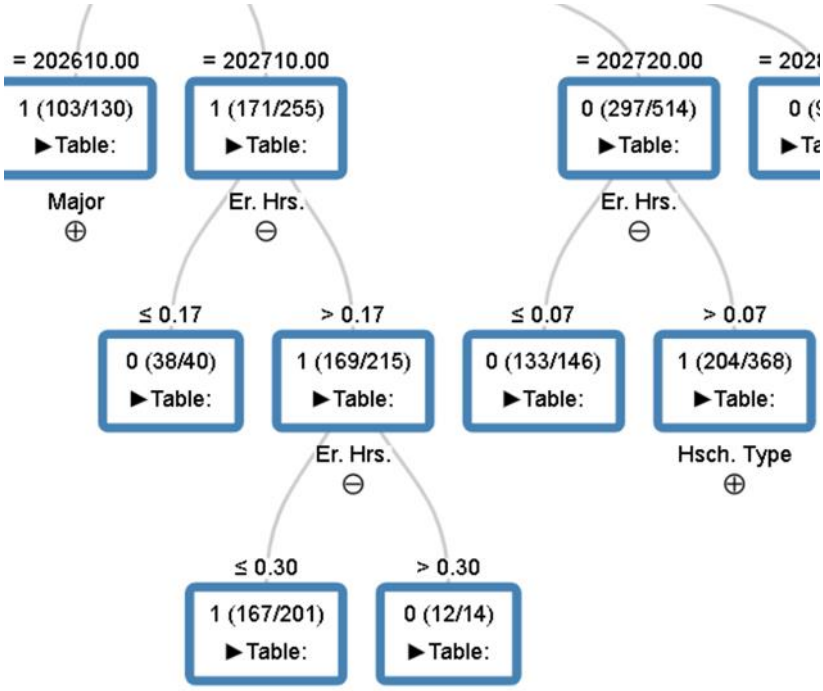
Figure 4: Decision Tree model



The main feature that was selected by the trained model to split the predictive steps is Expected Graduation Term. As shown in Figure 4, most of the students that are expected to graduate in 202820 and 202810 in the first partition of the data were at risk. On the other hand, most of the students who had the following expected graduation terms: 202510, 202520, 202420, 202620, were not at risk.

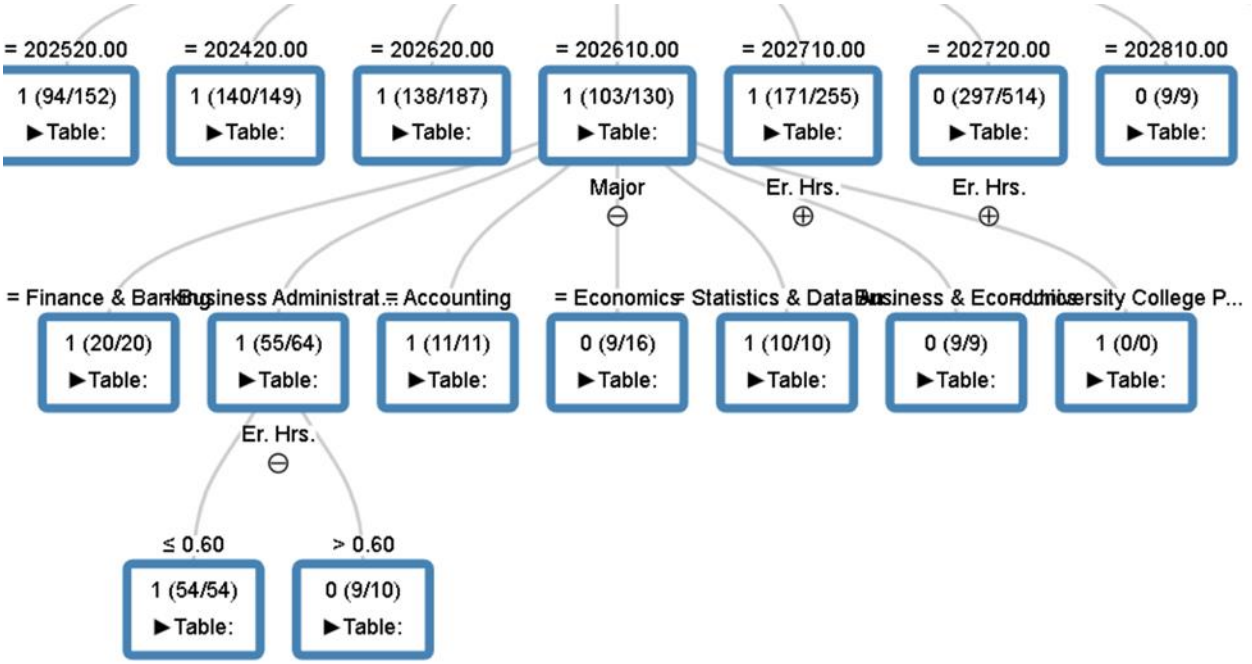
For the rest of expected graduation terms, a second feature was needed to be more decisive for these 3 terms: 202610, 202710, 202720.

Figure 5: second and third branches of the decision tree.



For the students with expected graduation terms of 202710 and 202720, the second feature was made by the independent variable Er. Hours. most of the students who are expected to graduate in 202710 and completed 17% or less of their credit hours were at risk, while the ones who completed between 17.01% and 30% of their credit hours were not at risk. While most of the ones who completed were than 30% of their credit hours were at risk. On the other hand, most of the students who were expected to graduate in 202720 and completed less than or equal to 7% of their credit hours were at risk.

Figure 6: second and third features in the decision tree.



For the students who are expected to graduate in 202610, major was used as a second feature to have more depth in the prediction. For the trained model, most of the students who are expected to graduate in 202610 and were studying Economics or Business & Economics are expected to be at risk, while for all other majors they were not at risk. University College Students were not considered because none of them are expected to graduate in 202610 in the first place.

Table 5: Confusion Matrix of the Decision Tree.

Actual / Predicted	Not At Risk (1)	At Risk (0)
Not At Risk (1)	402	27
At Risk (0)	20	26

In our case, 0 (student at risk) is Positive while 1 (student not at risk) is Negative.

Accuracy = (TP + TN) / (TP + TN + FP + FN) = (26+402) / (26+402+27+20) = 0.901 = 90.1%

Recall = TP/(TP+FN) = 26 / (26+20) =0.565 = 56.5%

Precision = TP/(TP+FP) =26 / (26+27) = 0.491 = 49.1%

Specificity = TN/(TN+FP) = 402 / (402+27) =0.937 = 93.7%

Figure 7: Accuracy of prediction for the decision tree mode.

Row ID	TruePo...	FalsePo...	TrueNe...	FalseN...	Recall	Precision	Sensitivity	Specificity	F-meas...	Accuracy	Cohen'...
1	402	20	26	27	0.937	0.953	0.937	0.565	0.945	?	?
0	26	27	402	20	0.565	0.491	0.565	0.937	0.525	?	?
Overall	?	?	?	?	?	?	?	?	?	0.901	0.47

Afterwards, we used the second partition of the data (30%) for testing the model. As we can see in the two Figures above, the model was able to predict 402 of the students who are not at risk correctly while it predicted 27 of them wrongly. It was also able to predict 26 of the students who are at risk correctly, while it predicted 20 of them wrongly. Overall, the accuracy of prediction for the decision tree is 90.1%, which means the error in prediction was 9.9%.

2. Random Forest:

Following up on the previously used machine learning mode, Decision Tree, we attempted to use the Random Forest model as an alternative way to predict the students at risk.

The Random Forest model is an ensemble machine learning model made up of several decision trees, thus creating a “forest” of them (IBM, n.d.). If the model was used for regression, then the results will be aggregated or averaged, while for classification models, the majority vote will decide the output.

The accuracy results achieved in the Random Forest, however, were less accurate in comparison at 88.2% as per the below table:

Table 6: Confusion Matrix of the Random Forest.

Actual / Predicted	Not At Risk (1)	At Risk (0)
Not At Risk (1)	391	38
At Risk (0)	18	28

The model correctly identified 419 records, while it was incorrect in identifying 56 records.

Accuracy = (TP + TN) / (TP + TN + FP + FN) = (28+391)/ (391 + 38 + 18 + 28) = 0.882 = 88.2%

Recall = TP/(TP+FN) = 28 / (28+18) =0.609 = 60.9%

Precision = TP/(TP+FP) =28 / (28+38) = 0.424 = 42.4%

Specificity = TN/(TN+FP) = 402 / (402+38) =0.911 = 91.1 %

Despite it being usually considered the more accurate model between Decision Trees and Random Forests, in this case it scored lower due to two potential causes: our small dataset or a lack of complexity in our data.

Random Forest are used to aggregate or classify many trees, which in turn means that it achieves better results using a larger number of records instead of a smaller number. Furthermore, because it usually works with many decision trees, the data it works with is both diverse and complex in comparison to the one we are working with. Which in turn means, that while it's not a bad model to use for this analysis, there are potentially more accurate ones to use.

3. Logistic Regression

3.1 Introduction

Logistic regression is a widely used statistical method for binary classification that models the relationship between one or more independent variables and a binary dependent variable. In the context of education, logistic regression can be an invaluable tool for identifying students at risk of academic failure. By leveraging data on student demographics, academic performance, and behavioral patterns, educational institutions can proactively address the needs of at-risk students, ultimately enhancing student success and retention.

3.2 Overview of Logistic Regression

Logistic regression estimates the probability that a given input point belongs to a particular category (e.g., at risk or not at risk). It uses the logistic function to map predicted values to a probability range between 0 and 1, allowing for easy interpretation of outcomes. (IBM, 2024)

3.3 Confusion Matrix Data Analysis

Table 6: Confusion Matrix of the Logistic Regression.

Actual / Predicted	Not At Risk (1)	At Risk (0)
Not At Risk (1)	371	12
At Risk (0)	27	54

1. True Positives (TP):
    - Correctly predicted "At Risk": 54
2. True Negatives (TN):
    - Correctly predicted "Not at Risk": 371
3. False Positives (FP):
    - Incorrectly predicted as "At Risk": 12
4. False Negatives (FN):

- Incorrectly predicted as "Not at Risk": 27

3.4 Analysis of Confusion Matrix

The confusion matrix provides a clear snapshot of the model's predictive performance:

- The model correctly identified 371 students as "Not at Risk," resulting in a high true negative rate.
- It successfully identified 54 students as "At Risk," but it also missed 27 students who were at risk, leading to false negatives.
- The false positive count was relatively low, at 12, indicating that when the model predicts a student as "At Risk," it is often correct.

3.5 Performance Metrics Summary

Key Metrics Analysis

- Recall (69.7%): The model captures about 69.7% of the at-risk students. While this is a solid level of recall, there is still a significant portion of at-risk students that are not being flagged.
- Precision (33.3%): A precision of 33.3% indicates that when the model predicts a student is "At Risk," it is correct only 33.3% of the time. This low precision suggests a high rate of false positives, meaning many flagged students may not require intervention.
- Specificity (87.3%): The model demonstrates good specificity at 87.3%, effectively identifying students who are "Not at Risk" and minimizing unnecessary alerts.
- Accuracy (85.8%): An overall accuracy of 85.8% reflects the model's strong performance in correctly classifying both at-risk and not-at-risk students. However, this number should be interpreted with caution in the context of class imbalance.

3.6 Conclusion

The analysis of the confusion matrix reinforces the logistic regression model's effectiveness in identifying at-risk students. While the model excels in specificity and overall accuracy, the recall metric highlights an area for improvement, as some at-risk students are not being flagged. This information is vital for institutions to focus their interventions effectively and ensure support reaches those in need.

4. Gradient Boosting Trees Model

4.1 Introduction about the Model:

Gradient Boosting Trees model is used for both classification (decision tree) and regression. The model is built to predict by combining the weaker learners into strong models and to minimize the residual errors (the difference between the actual value and the predicted value) from previous models which allow the models working in highly efficient ways to capture the complex patterns in data (Fafalios et al., 2020). The model aims to minimize a loss function L(y,y^) (where:

- y = true value (actual target variable)
- y^ = predicted value by the model

The model works by taking the first prediction then calculating the residual (error) by using following formula ri =yi −y^ i, where ri is the residual/error for the *i.* Then the model tries to fit A new decision tree (weak learner) if it fit to the residuals, it would mean the tree tries to predict the errors made by using the same previous process. Then the prediction will be updated by adding the prediction of the new tree. If the prediction from the tree is *f(x)*, the new prediction becomes:

$\hat{y}_{new} = \hat{y}_{old} + v f(x)$

where v is the learning rate (a small number, like 0.1, to help controls the correction each new tree adds). Then the model will repeat the process by building more trees which will be corrected by using the previous model's errors, until the errors (residuals) are minimized.

4.2 Explanations of the Model process in KNIME

The process followed the same preparation process which was used on the other models. The purpose of using the same process is to have a clear comparison between the models' results. The model used the GPA Class to predicate the significant factors that contribute to having the students at risks.

Through KNIME, the partition node has been used to split the dataset into two parts. The first part is for training the dataset and the second part to test the remining model. The configuration of the node selected to be 70% to be used for training the model and the remining 30% to be used for testing the model.

As the dataset has imbalanced data, the SMOTE node was used to link the data in the partitioning node. The node is used to ensure the data is not biased or overfitting which helps the model balance the data and better learn the characteristics.

Through the SMOTE node we link the Gradient Boosting Trees learning. The model is used to assess the effectiveness of the model in identifying the most significant factors that contribute to having the students at risk and to comparing the evaluation model with the other models. The configuration of the Gradient Boosting Trees model learning selects the GPA class as the target to predicate with all attributes.

4.3 Results of the Model

The connection from partitioning and Gradient Boosting Trees learner, Gradient Boosting Trees predictor will be use. The results of the model as the following:

Table 7: Confusion Matrix of the Gradient Boosting Model.

Actual / Predicted	1 (Not at Risk)	0 (at Risk)

1 (Not at Risk)	411 (TN)	18 (FN)
0 (At Risk)	21 (FP)	25 (TP)
Model Correct classified 436,		
Model wrong classified 39		
Model Accuracy 91.789%		
Model Error 8.211%		
Recall: TP/TP+FN=25/ (25+18) =0.58		
Precision: TP/TP+FP=25(25+21) = 0.50		
Specificity: TN/(TN+FP) =411(411+21) =)0.95		

Accuracy and Confusion Matrix Results:

The model correctly classified 436 students (411students not at risk and 25 students at risk) and the model incorrectly classified 39 students (18 students are classified as not at risk and 21 students were classified as at risk).

Accuracy Matrix Results:

- The model has very high accuracy results (91.789%) which means that the model works effectively to predicate the students at risk or not with a low error percentage (8.211%).

Confusion Matrix Results:

- Ture positive (TP): the model classified 25 students are at risk correctly.
- True Negatives (TN): the model classified 411 students are not at risk correctly.
- False Positives (FN): the model classified 18 students are at risk incorrectly
- False Negatives (FP): the model classified 21 students are not at risk incorrectly

Recall Results: Recall is the ability of the model to identify correctly the False Positives (At-Risk students). As shown from the results, the model has classified 58% of all the students at risk.

Precision Results: Precision is the ability of the model to indicate and predict correctly the False Negatives and the results are 50%.

Specificity Results: Specificity is the ability of the model to identify students not at risk (True Negative) and the results are 95% of the model's identification.

Overall results: the model achieves high scores in accuracy and in specificity, while the precision and recall scores are normal. Which means that the model performs well in identifying **the students at risk (accuracy percentage) and those not at risk**. While

also performing adequately in false negative and false positive of students at risk due to the nature of the dataset which include a low number of the students at risk.

5. Support Vector Machine Model:

The Support Vector Machine (SVM) is a supervised learning algorithm. Vapnik introduced it as a kernel based on classifications and regression tasks (Shmilovici, 2005). It is a powerful algorithm mainly used for binary classification and regression analysis. When identifying precise patterns in complicated datasets, SVM is far more adept than other machine-learning techniques, which are used to recognize handwriting, speaker identification, and faces and detect fraudulent credit cards (Huang d al., 2018). In addition, SVM is commonly used in the medical field. It primarily recognizes cancer patterns, outcome prognosis, drug benefit prediction, or a tumor-specific biological process (Huang et al., 2018).

In SVMs, a key concept is the 'hyperplane ', which is a decision boundary between two classes. This hyperplane is oriented to be as far away as feasible from the nearest data points for each class. These nearest points are known as support vectors. The primary goal of training an SVM model is to find the weight vector and the bias so that they hyperplane divide the data and maximize the margin in the data (Huang et al., 2018).

One advantage of the SVM model is the ability to detect complex patterns from big datasets. Also, the SVM can receive a subset of support vectors during the learning phase, which is frequently only a tiny part of the original data set. A tiny data collection is used to create this set of support vectors representing a particular classification task. However, SVMs have some limitations. The process of selecting the parameters is complex, which could affect the classifier's training time in large data sets. Also, the development of optimal classifiers for multi-class problems is very complex. Furthermore, the performance of SVM can be affected by the problem of imbalanced classified data sets (Shmilovici, 2005).

Table 8: Confusion Matrix for SVM Model.

Actual / Predicted	Not At Risk (1)	At Risk (0)
Not At Risk (1)	429	0
At Risk (0)	46	0

After applying the SVM model to our dataset, we found that the model accuracy was 90.3%, which is high. However, the confusion matrix shows that the model has classified all the training datasets, 475 records, as students above risk. This could be due to the imbalanced classification in the dataset.

As mentioned, the SVM is sensitive and performs lower on the imbalanced dataset. The SVM seems to produce a model biased towards the majority class and performed poorly on the minority class. As we mentioned before, during the preparation phase, our dataset has imbalanced classifications. We tried to solve this issue by using the SMOTE node, but the model still detects bias in the dataset. And that occurs because imbalance

data causes an imbalance ration between the positive and negative support vectors, so the model predicts the most frequented class (Batuwita, R. & Palade, V, n.d.).

Discussion

This study aims to identify the characteristics of at-risk students and develop a robust prediction model using machine learning algorithms. The percentage of at-risk students at UAEU, particularly in the Business & Economics College, has seen a worrying increase in recent years. This trend has rightly drawn the attention of decision-makers, as it has the potential to significantly impact student performance. Our analysis has brought to light the most significant insights into this pressing issue.

For this analysis, we meticulously collected data from the Fall 2023 semester, comprising 1729 records. We then classified the students into two classes based on their cumulative GPA: 1 for those above the risk threshold (Cumulative GPA above 2) and 0 for those at risk (Cumulative GPA equal to or less than 2).

After cleaning and processing the data, we found that the at-risk students represent 10% of the Business & Economics College students, which is considered an issue that could increase over time. The visual analysis found that most at-risk students are Emirati females, which is reasonable as the majority of the students in the University are females. 68% of students at risk are from the Abu Dhabi Emirate. We have noticed that 71% of the students came from government schools. In addition, 40% of the students majoring in Business Administration are at risk with a GPA less than 2, and 36% are majoring in Business & Economics. Moreover, 46% of the students are new students with credit hours equal to or less than 15.

Having identified the common characteristics of at-risk students, we then applied a range of advanced machine-learning models. Our goal was to determine the most accurate model for classifying at-risk students and predicting the future student population of the UAEU Business & Economics College.

Table 9: Comparison between ML Models.

Model	Random Forester	Logistic Regression	Decision Tree	Support Vector Machine	Gradient Boosted
Accuracy	**88.2%**	**85.6%**	**90.7%**	**90.3%**	**91.4%**

The above table shows that the Support Vector Machine, Gradient Boosted, and the Decision Tree have the highest accuracy. But the Gradient Boosted model has the highest accuracy overall. However, we decided to exclude SVM because we believe it’s affected by the imbalance in the dataset. The SVM classified the tested dataset into one

class (Above Risk). However, we used the SMOTE node in KNIME software to deal with issues, smooth data, and balance it.

So, to avoid bias, we recommend using the Gradient Boosted model to predict class for the new students. The accuracy of the model is 91.4%, which is the highest. It is reasonable to use the Gradient-Boosted model as it predicted the new observations by combining different learners into a robust model and minimizing the residual errors, which in turn, creates a more efficient model that detects the complex patterns in the data.

Conclusion

In Conclusion, our research investigates the at-risk students at UAE University, especially the Business and Economics College. We found that if a new student is an Emirati female student from Abu Dhabi and has credit hours of less than 15 then she is more likely to be at risk. Government high school students are more likely to be at risk than other students. Business Administration or Business & Economics students are more likely to be at risk. The Gradient Boosting algorithm is the most suitable model for predicting at-risk students, with an accuracy of 91.4%.

Meanwhile, Logistic Regression and Random Forest have lower accuracy than the other models. That is unassociated with the finding in Latrellis et al. (2021). Overall, the ability to apply the Machine learning algorithm to predict students' status (at risk or above risk) is a very effective strategy for the university to limit the increasing risk of having students at risk.

1. Future Recommendations

One of the best practices that will limit the student-at-risk phenomenon in the university is introducing a club or hub for those students. The club will target at-risk students and provide all the support they need. Daily academic and life skills workshops will be provided to improve students' skills. It will allow the students to share their difficulties and take advantage of other students' experiences. In addition, using the ML algorithm to detect students at risk early and include them in the club will reduce the difficulty of solving the issue of students at risk. Another practice is to allow the student at risk to adjust their course load according to their ability. That will help them focus more on improvement areas and maintain their graduation schedule.

2. Research Limitations

One of the research limitations is the imbalanced dataset. The imbalanced classification of the data causes bias, especially when using prediction models. We used the resampling method (SMOTE technique) to create a balanced dataset in the data preparation process. However, one of the models was detected to provide bias prediction despite our usage of the node. Therefore, applying another method would

help us avoid this problem. The ensemble learning method can be used, where the majority class dataset is divided into several smaller sub-datasets, each with a comparable quantity. Additionally, SVMs could be used with boosting algorithms like Adacost, RareBoost, and SMOTEBoost, which can be applied in class imbalance learning with ensemble settings. (Batuwita, R. & Palade, V, n.d.).

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